LinFill-120 Vertical Syringe Filling Station

LinFill Series · Elara Automation GmbH

Lifecycle Information Document

The LinFill-120 is a compact, vertically-actuated syringe filling station designed for small-batch pharmaceutical and laboratory filling applications. This lifecycle document covers the full operational lifespan of the product: from pre-installation planning through routine operation, maintenance, and eventual decommissioning.

It should be read in conjunction with the MQTT Interface Specification (EA-MI-LF120-EN) and any site-specific installation qualification documentation prepared by the equipment owner.

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About This Document

This document has been prepared by Elara Automation GmbH and is intended for qualified technical personnel responsible for the installation, commissioning, operation, and maintenance of the equipment described herein. Readers are expected to be familiar with general principles of industrial automation, electrical safety, and network-connected laboratory equipment.

The document is structured as a reference manual. Sections are broadly ordered to follow the typical lifecycle of the equipment — from first delivery through to end-of-life disposal — but individual sections may be read in isolation. Cross-references are provided where information in one section depends on material covered elsewhere.

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Conventions Used in This Document

The following conventions are used throughout:

Notation	Meaning
{serialNumber}	The unit's serial number as printed on the rear label (e.g. EA-PS-2024-00391)
{base}	The site-specific topic or namespace prefix, configured during commissioning
NOTE	Supplementary information that aids understanding but is not safety-critical
CAUTION	Risk of equipment damage or process disruption if instruction is not followed
WARNING	Risk of personal injury if instruction is not followed

Related Documents

Document No.	Title
EA-MI-LF120-EN	LinFill-120 MQTT Interface Specification
EA-LN-SPL01-EN	SPL-01 Line Description & Bill of Materials
EA-LC-PS080-EN	PlungerSet-80 Lifecycle Information Document

1 Product Overview

The LinFill-120 is intended for laboratory and pilot production use in small-batch pharmaceutical filling lines. The station accepts a removable needle top compatible with Sanisure Fill4Sure and Cytiva Allegro-style needles. Vertical positioning is achieved by a 12 V DC motor driving an 8 mm lead screw with 2 mm pitch, giving 150 mm of travel. Two integrated limit switches provide homing detection and upper travel-limit protection.

The onboard ESP32-WROOM-32 microcontroller manages motion control and exposes the station's state and data over WiFi using MQTT 3.1.1, making it directly compatible with Unified Namespace architectures. The firmware is shared with the PlungerSet-80 stoppering station; both products use the same base platform, motor driver board, and lead-screw assembly.

The LinFill-120 is **not** certified for use in classified pharmaceutical clean rooms above ISO Class 7 without additional enclosure measures. It is not a safety device and must not be relied upon for safety-critical functions.

Typical Application

In a typical pilot filling line the LinFill-120 is positioned as the first active station. A pre-filled syringe is placed manually in a carrier tray beneath the needle top. Upon receipt of a START command the lift descends, the filling stroke is executed, and the lift retracts. The station then publishes the cycle time and — if an external balance is connected — a post-fill syringe weight reading. The filled syringe is then passed to a downstream stoppering station for plunger insertion.

2 Delivery and Storage

On receipt, inspect the outer packaging for visible transit damage before opening. The carton contains: the LinFill-120 station (1×), the AC/DC power adapter appropriate for the ordered region (1×), M6 mounting hardware (1 set), and a printed quick-start card. The needle top and needle consumables are not included and must be ordered separately.

If the unit will not be put into service immediately it should be stored in its original packaging in a dry, dust-free environment. Acceptable storage temperatures range from –20 °C to +60 °C. Prolonged storage above 40 °C or below 0 °C is not recommended for the 3D-printed ABS components of the upper assembly.

Record the serial number — printed on the rear label and on the packing slip — in the site asset register before installation. The serial number is required when requesting support, ordering spare parts, or applying firmware updates.

NOTE: *The serial number format is EA-LF-YYYY-NNNNN (e.g. EA-LF-2024-00472). Year of construction is encoded in the serial number.*

3 Installation

Installation must be carried out by qualified personnel familiar with both the mechanical mounting requirements and the network configuration of the facility. The following subsections cover mechanical, electrical, and network installation in the recommended sequence.

3.1 Mechanical Mounting

The LinFill-120 mounts on a flat surface using four M6 bolts on a 150 × 120 mm bolt-circle pattern. The mounting footprint is compatible with the ACOPOS table mounting surface and similar laboratory automation platforms. The table surface must be level to within ±0.5 mm across the mounting footprint; shim if necessary.

Allow a minimum of 100 mm clear headroom above the needle top in the fully-extended position. With the needle retracted, the overall height is approximately 310 mm; with needle extended, approximately 450 mm. Width and depth are 220 mm × 180 mm respectively. Unit weight is approximately 2.4 kg without needle consumable.

CAUTION: The lead screw and motor assembly contains moving parts. Keep hands clear of the vertical travel path at all times during operation. Do not place fingers beneath the needle top while the station is powered.

3.2 Electrical Connection

Connect the included power adapter to the 5.5/2.1 mm barrel jack on the rear of the unit (centre positive, 12 V DC). The adapter accepts 100–240 V AC at 50/60 Hz and is rated for the region specified in the order code (EA-LF120-230V-EU for European mains). Do not use a power supply rated below 3 A; the unit draws up to 18 W during full actuation.

A 3.15 A slow-blow fuse (F1) is accessible on the rear panel. Replace only with an identically rated fuse. Fitting a higher-rated fuse will void the warranty and may create a fire hazard. The ESP32 module is powered from an onboard 3.3 V LDO regulator; the motor controller is powered from the 12 V rail switched by the L298N H-bridge.

3.3 Network Provisioning

On first power-on the ESP32 broadcasts a provisioning access point named "LinFill-Setup". Connect a laptop or mobile device to this network and navigate to 192.168.4.1 to open the provisioning web interface. Enter the facility WiFi SSID, passphrase, and MQTT broker address. The device will reboot and connect. Assign a static IP or DHCP reservation to ensure the broker can reach the device consistently.

The client identifier used on the MQTT broker is formed from the product family prefix and the serial number. For this product family the prefix is *linfill120-*. The keepalive interval is fixed at 60 seconds in firmware v1.4.2. Details of the MQTT topic structure, payload formats, and security recommendations are covered in the interface specification (EA-MI-LF120-EN).

4 Commissioning and Qualification

Before routine production use, the station must be commissioned and — where required by site quality procedures — subjected to Installation Qualification (IQ) and Operational Qualification (OQ). The commissioning steps below are the minimum recommended by Elara Automation; sites with more stringent requirements should supplement these with their own protocol.

4.1 Commissioning Steps

Step	Action	Acceptance Criterion
C-01	Power on; confirm power LED illuminates	LED steady green within 5 s
C-02	Confirm WiFi connection on provisioning interface	IP address assigned; RSSI ≥ –70 dBm
C-03	Subscribe to {base}/filling/state on MQTT broker	"IDLE" or "ERROR" received within 10 s of power-on
C-04	Publish HOME command to {base}/filling/cmd	State transitions IDLE → HOMING → IDLE; homing cycle completes without fault
C-05	Publish START command; load a blank syringe	State transitions IDLE → RUNNING → IDLE; cycletime published
C-06	Publish STOP during an active cycle	State returns to IDLE; lift retracts to safe position
C-07	Induce an error state; publish RESET	ErrorCode clears; State returns to IDLE
C-08	Verify syringe weight topic (if balance connected)	{base}/filling/syringeweight published after cycle completion

4.2 Lift Repeatability Verification

Positional repeatability of the lift (specified as ±0.5 mm after a homing cycle) should be verified during OQ using a digital indicator mounted to the frame with the probe contacting the needle-top carriage. Issue three HOME commands in sequence and record the indicator reading after each homing cycle completes. The range of the three readings must not exceed 1.0 mm. Repeat if the limit switch is replaced.

5 Normal Operation

During normal operation the LinFill-120 accepts four commands — START, STOP, HOME, and RESET — published to its command topic by a supervisory SCADA, MES, or manual operator interface. The station publishes its state, cycle time, and syringe weight to the MQTT broker in response to operational events.

The station does not have a local HMI; all control and monitoring is performed via the network interface. Integrators should ensure that the supervisory system monitors the state topic continuously and handles ERROR states by alerting the operator and inhibiting further START commands until a RESET has been acknowledged.

NOTE: *Allow a minimum of 5 seconds between consecutive START commands to allow the lift to return fully to the home position and the balance reading to stabilise.*

5.1 State Machine Summary

The station operates as a four-state machine. IDLE is the normal resting state following a successful HOME or the completion of a fill cycle. RUNNING indicates a cycle is in progress; no further commands are accepted until the cycle completes or a STOP command is received. HOMING is entered when a HOME command is received from any state; it is also the initial state on power-on if no auto-home is configured. ERROR latches when a fault condition is detected and inhibits START until RESET is issued. See the interface specification for the full state-transition table.

6 Preventive Maintenance

The maintenance interval recommended by Elara Automation is every 500 operating hours or at least once per year, whichever comes first. All maintenance activities should be logged in the site maintenance record with date, technician name, and observations.

Before carrying out any maintenance that requires access to moving parts, isolate the station from its power supply by removing the barrel-jack connector. Do not rely solely on the STOP command or a powered-off state from the MQTT broker.

WARNING: Always isolate power before accessing the lead screw, motor, or limit-switch assemblies. The 12 V motor supply is present at the L298N board terminals even when the ESP32 firmware is not commanding motion.

6.1 Scheduled Tasks

Task	Procedure	Acceptance
Lead-screw lubrication	Apply a small amount of NLGI #2 lithium grease along the full length of the lead screw. Manually traverse the carriage to distribute the lubricant. Wipe excess from the screw and guide rails.	Screw moves freely; no binding or stiction
Limit-switch inspection	With power isolated, manually depress each lever arm and confirm snap-action feel. With power applied, issue a HOME command and confirm the homing cycle completes without HOMING_TIMEOUT.	Both switches actuate positively; HOME cycle completes in < 15 s

Task	Procedure	Acceptance
Fastener check	Check all M3 screws on the motor mount, switch mounts, and guide-rail brackets for tightness. Use a torque driver set to 0.5 N·m.	No loose fasteners
Needle top inspection	Remove the needle top assembly and inspect the clip mechanism for wear or contamination. Clean with IPA wipe if required. Replace if clips do not engage positively (order code EA-LF120-SPARE-TOP).	Needle top seats fully and does not rotate under hand torque
WiFi / broker check	Confirm the station is connected to the broker and publishing state messages. Check RSSI from the provisioning web interface; reposition the access point if RSSI is below –70 dBm.	RSSI ≥ –70 dBm; MQTT state received within 5 s of power-on

7 Firmware Updates

Firmware for the LinFill-120 and PlungerSet-80 is developed on a shared platform codebase. Update files (.bin) are distributed by Elara Automation support and should only be applied to the specific product variant for which they are issued. Applying a PlungerSet-80 firmware file to a LinFill-120 unit will result in incorrect behaviour and will not be covered by warranty.

Updates are applied over-the-air via the device web interface. The station does not need to be removed from service permanently; the update takes approximately 45 seconds, after which the device reboots automatically. Plan updates to coincide with scheduled downtime where possible, as the station will be unavailable for approximately two minutes during the update and reboot cycle.

7.1 Update Procedure

- Connect the host computer to the same WiFi network as the station.
- Open a browser and navigate to **http://linfill120-{serialNumber}.local**
- Log in with the administrator credentials set during provisioning.
- Select **System > Firmware Update** from the navigation menu.
- Click **Choose file**, select the .bin file supplied by Elara Automation, then click **Upload & Apply**.
- Do **not** close the browser or power off the unit while the progress bar is visible.
- After the device reboots, confirm the firmware version displayed on the System page matches the expected version.
- Issue a HOME command and verify the station returns to IDLE before resuming production.

7.2 Firmware Version History

Version	Release	Summary of Changes
1.4.2	2024-03-01	Fixed rare watchdog reset during extended HOMING cycles. Improved MQTT reconnection logic after broker unavailability. Syringe weight averaged over 3 readings (previously 1 reading).
1.4.1	2023-11-15	Added QoS Level 1 publish retry queue for messages sent during transient broker disconnection.
1.4.0	2023-08-20	Initial production release for the LinFill-120 / PlungerSet-80 shared platform.

8 Spare Parts and Accessories

Genuine Elara Automation spare parts should always be used when replacing components. Use of non-genuine parts may affect performance, void the warranty, and compromise the CE declaration of conformity. Parts can be ordered through the Elara Automation online store or by contacting the support team using the details on the back page.

Order Code	Description	Notes
EA-LF120-SPARE-TOP	Removable needle top assembly (without needle)	User-replaceable; no tools required
EA-LF120-SPARE-MTR	Replacement 12 V DC motor assembly with lead-screw coupling	Service personnel only; requalification required after replacement
EA-LF120-SVC-1Y	1-year extended service contract	Includes annual preventive maintenance visit and 48 h on-site response

Needle consumables — Sanisure Fill4Sure and Cytiva Allegro Style — are not supplied by Elara Automation and must be ordered from the respective manufacturer. Record needle lot numbers in the batch record.

9 Calibration and Requalification

The LinFill-120 does not contain internally calibrated metrology instruments. The syringe weight value published after each fill cycle is derived from an external balance connected to the ESP32 UART; calibration of that balance is outside the scope of this document and is the responsibility of the equipment owner.

The lift positional repeatability (±0.5 mm after homing) should be re-verified following any of the events listed below. Where site quality procedures require formal requalification, repeat the OQ protocol prepared during initial commissioning.

- Replacement of the lead screw assembly or DC drive motor
- Replacement of either limit switch
- Any change to the physical mounting location or table surface
- A firmware update (minimum: functional verification of all four commands)
- Any impact or mechanical shock to the unit

10 Troubleshooting

Symptom	Probable Cause	Corrective Action
State stuck in HOMING for > 20 s	Homing limit switch not reached within timeout; switch failure or obstruction	Check for mechanical obstruction in lift travel. Verify limit switch continuity (5 A, snap-action). If switch faulty, replace and requalify.
State transitions to ERROR immediately after START	Motor stall or L298N thermal shutdown	Check for mechanical binding on lead screw. Allow L298N to cool (2 min). Check 12 V supply voltage; must be ≥11.4 V under load.
{base}/filling/syringeweight not published	Balance not connected or UART misconfigured	Check UART wiring from balance to ESP32. Verify baud rate in provisioning UI.
Station not visible on broker after power-on	WiFi association failure or DHCP lease expired	Re-open provisioning AP (hold reset button 5 s), re-enter WiFi credentials. Assign static DHCP lease on router.

Symptom	Probable Cause	Corrective Action
Cycletime values significantly longer than baseline	Increased lead-screw friction	Inspect and lubricate lead screw per §6. Check for ABS debris in guide channel.

11 Decommissioning and Disposal

When the LinFill-120 reaches end of operational life it must be decommissioned safely before disposal. The decommissioning procedure below ensures that no residual process materials remain in the station and that data security obligations are met.

- Issue a HOME command and confirm the lift returns to the home position. Power off the station from the supply.
- Remove and dispose of the needle top assembly and any fitted needle consumable according to site procedures for pharmaceutical-contact waste.
- Erase WiFi credentials and MQTT broker configuration by performing a factory reset (hold the rear-panel reset button for 10 seconds until the LED flashes amber).
- Remove the M6 mounting bolts and lift the station clear of the work surface.
- The powder-coated steel base is recyclable as ferrous metal. The ESP32, L298N, and associated PCBs must be disposed of as WEEE (Directive 2012/19/EU). Contact a certified WEEE recycler.
- Issue a decommissioning record noting the serial number, date, and disposal route.

12 Certifications and Regulatory Compliance

The LinFill-120 bears the CE marking in accordance with the following EU directives:

- Low Voltage Directive (LVD) 2014/35/EU
- EMC Directive 2014/30/EU
- RoHS Directive 2011/65/EU — restriction of hazardous substances in electrical equipment

The integrated WiFi radio (ESP32-WROOM-32 module) carries FCC ID 2AC7Z-ESPWROOM32.

IP rating: IP20. The unit is rated for Pollution Degree 2 and Installation Category II in accordance with IEC 60664-1. It is intended for indoor use only.

13 Warranty

Elara Automation GmbH warrants the LinFill-120 against defects in materials and workmanship for a period of 24 months from the date of shipment. This warranty does not cover consumable items (needle tops, needle consumables), damage to the ESP32 WiFi antenna caused by physical impact, or damage resulting from operation outside the electrical or environmental specifications stated in this document.

Warranty claims should be submitted to Elara Automation GmbH support within the warranty period, quoting the unit serial number, date of purchase, and a description of the defect. Elara Automation GmbH reserves the right to repair or replace the unit at its sole discretion. Unauthorised disassembly of the unit will void the warranty.

Manufacturer & Technical Support	Sales & Order Enquiries
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